

```
tusharkapoor@LAPTOP-PI08CKLL:/mnt/c/Users/Sukesh$ whoami
```

```
tusharkapoor
```

```
tusharkapoor@LAPTOP-PI08CKLL:/mnt/c/Users/Sukesh$ cd ~
```

```
pwd
```

```
/home/tusharkapoor
```

```
tusharkapoor@LAPTOP-PI08CKLL:~$ cat /etc/os-release
```

```
PRETTY_NAME="Ubuntu 24.04.4 LTS"
```

```
NAME="Ubuntu"
```

```
VERSION_ID="24.04"
```

```
VERSION="24.04.4 LTS (Noble Numbat)"
```

```
VERSION_CODENAME=noble
```

```
ID=ubuntu
```

```
ID_LIKE=debian
```

```
HOME_URL="https://www.ubuntu.com/"
```

```
SUPPORT_URL="https://help.ubuntu.com/"
```

```
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
```

```
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
```

```
UBUNTU_CODENAME=noble
```

```
LOGO=ubuntu-logo
```

```
tusharkapoor@LAPTOP-PI08CKLL:~$ sudo apt update
```

```
[sudo] password for tusharkapoor:
```

```
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
```

```
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
```

```
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
```

```
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
```

```
Reading package lists... Done
```

```
Building dependency tree... Done
```

```
Reading state information... Done
```

```
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
tusharkapoor@LAPTOP-PI08CKLL:~$ sudo apt upgrade -y
```

```
Reading package lists... Done
```

```
Building dependency tree... Done
```

```
Reading state information... Done
```

```
Successfully installed contourpy-1.3.3 cyclor-0.12.1 fonttools-4.63.0 kiwisolver-1.5.0 matplotlib-3.11.1 numpy-2.5.2 packaging-26.3 p
andas-3.0.5 pillow-12.3.0 psutil-7.2.2 pyparsing-3.3.2 python-dateutil-2.9.0.post0 six-1.17.0
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ python -c "import psutil, pandas, matplotlib; print('PYTHON LAB ENVIRON
MENT READY!')"
PYTHON LAB ENVIRONMENT READY
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ touch README.md requirements.txt .gitignore
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ python -m pip freeze > requirements.txt
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ ls -la
total 16
drwxr-xr-x 3 tusharkapoor tusharkapoor 4096 Aug 20 07:31 .
drwxr-x--- 8 tusharkapoor tusharkapoor 4096 Aug 20 07:30 ..
-rw-r--r-- 1 tusharkapoor tusharkapoor 0 Aug 20 07:31 .gitignore
drwxr-xr-x 6 tusharkapoor tusharkapoor 4096 Aug 20 07:29 .venv
-rw-r--r-- 1 tusharkapoor tusharkapoor 0 Aug 20 07:31 README.md
-rw-r--r-- 1 tusharkapoor tusharkapoor 217 Aug 20 07:31 requirements.txt
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ mkdir -p common/{templates,test-data,screenshots,reports}

mkdir -p Lab1/Q1_System_Calls/{src,input,output,screenshots,tests}
mkdir -p Lab1/Q2_FCFS_SJF/{src,input,output,screenshots,tests}
mkdir -p Lab1/Q3_Priority_Round_Robin/{src,input,output,screenshots,tests}
mkdir -p Lab1/Q4_Metrics_Threads_IPC/{src,input,output,screenshots,tests}

mkdir -p Lab2/Q1_Thread_Models/{src,input,output,screenshots,tests}
mkdir -p Lab2/Q2_Race_Mutex_Semaphore/{src,input,output,screenshots,tests}
mkdir -p Lab2/Q3_Classical_Synchronization/{src,input,output,screenshots,tests}
mkdir -p Lab2/Q4_Deadlock_Banker/{src,input,output,screenshots,tests}

mkdir -p Lab3/Q1_Contiguous_Swapping/{src,input,output,screenshots,tests}
mkdir -p Lab3/Q2_Paging_Segmentation/{src,input,output,screenshots,tests}
mkdir -p Lab3/Q3_Page_Replacement/{src,input,output,screenshots,tests}
mkdir -p Lab3/Q4_Device_Management/{src,input,output,screenshots,tests}
mkdir -p Lab3/Q5_Disk_Scheduling/{src,input,output,screenshots,tests}
```

```
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
tusharkapoor@LAPTOP-PI08CKLL:~$ sudo apt install -y python3 python3-pip python3-venv git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3-pip is already the newest version (24.0+dfsg-1ubuntu1.3).
python3-venv is already the newest version (3.12.3-0ubuntu2.1).
git is already the newest version (1:2.43.0-1ubuntu7.3).
htop is already the newest version (3.3.0-4build1).
procps is already the newest version (2:4.0.4-4ubuntu3.2).
sysstat is already the newest version (12.6.1-2).
tree is already the newest version (2.1.1-2ubuntu3.24.04.2).
curl is already the newest version (8.5.0-2ubuntu10.12).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
tusharkapoor@LAPTOP-PI08CKLL:~$ python3 --version
pip3 --version
python3 -m venv --help
git --version
bash --version
htop --version
tree --version
iostat -V
curl --version
Python 3.12.3
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
usage: venv [-h] [--system-site-packages]
            [--symlinks | --copies] [--clear] [--upgrade]
            [--without-pip] [--prompt PROMPT] [--upgrade-deps]
            ENV_DIR [ENV_DIR ...]

Creates virtual Python environments in one or more target
```

Capstone/README.md
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab\$ tree -a -L 4 -I '.venv|__pycache__'

```
.
├── .gitignore
├── Capstone
│   ├── README.md
│   ├── data
│   ├── logs
│   ├── outputs
│   ├── screenshots
│   ├── src
│   └── tests
├── Lab1
│   ├── Q1_System_Calls
│   │   ├── README.md
│   │   ├── input
│   │   ├── output
│   │   ├── screenshots
│   │   ├── src
│   │   └── tests
│   ├── Q2_FCFS_SJF
│   │   ├── README.md
│   │   ├── input
│   │   ├── output
│   │   ├── screenshots
│   │   ├── src
│   │   └── tests
│   └── Q3_Priority_Round_Robin
│       ├── README.md
│       ├── input
│       ├── output
│       ├── screenshots
│       └── src
```

```
└─ tests
Lab3
├─ Q1_Contiguous_Swapping
│  ├── README.md
│  ├── input
│  ├── output
│  ├── screenshots
│  ├── src
│  └── tests
├─ Q2_Paging_Segmentation
│  ├── README.md
│  ├── input
│  ├── output
│  ├── screenshots
│  ├── src
│  └── tests
├─ Q3_Page_Replacement
│  ├── README.md
│  ├── input
│  ├── output
│  ├── screenshots
│  ├── src
│  └── tests
├─ Q4_Device_Management
│  ├── README.md
│  ├── input
│  ├── output
│  ├── screenshots
│  ├── src
│  └── tests
└─ Q5_Disk_Scheduling
   ├── README.md
   └── input
```

```
├── screenshots
├── src
├── tests
├── Q4_Free_Space_Permissions
│   ├── README.md
│   ├── input
│   ├── output
│   ├── screenshots
│   ├── src
│   └── tests
├── README.md
├── common
│   ├── reports
│   ├── screenshots
│   ├── templates
│   └── test-data
└── requirements.txt
```

119 directories, 21 files

```
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ python -c "import os, sys, psutil, pandas, matplotlib; print('OS:', os.name); print('Python:', sys.version.split()[0]); print('CPU:', psutil.cpu_count()); print('Memory:', psutil.virtual_memory().total); print('OPERATING SYSTEMS LAB SETUP READY')"
```

OS: posix

Python: 3.12.3

CPU: 12

Memory: 8154398720

OPERATING SYSTEMS LAB SETUP READY

```
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ code .
```

```
(.venv) tusharkapoor@LAPTOP-PIO8CKLL:~/Operating-Systems-Lab$ |
```



